Dasari Johan Abhishek

AI Software Engineer

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SUMMARY

AI Software Engineer experienced in building developer tooling, agentic systems, cloud-backed AI pipelines, and full-stack products. Creator of **ProjectBrain (projectbrain-mcp)** on PyPI—a local-first AST context compiler and static contract verifier for AI coding agents—with production experience across **Google Cloud Platform (GCP)**, **Model Context Protocol (MCP)**, database infrastructure, and TypeScript/Python engineering.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++.

AI & Developer Tooling: Model Context Protocol (MCP), Tree-sitter AST Analysis, Context Engineering, LangGraph, LangChain, PyTorch, Evals.

Backend & Frontend: FastAPI, Node.js, React, Next.js, REST APIs, OAuth2, Webhook & Third-Party API Integrations.

Cloud, Data & Infra: Google Cloud Platform (GCP), Cloud Storage (GCS), Firestore, PostgreSQL, SQLite (WAL), Docker, Vercel, Git, CI/CD.

EXPERIENCE

AI Software Engineer Intern **May 2026 - Present**

Saral Vidhya Hyderabad, India

- Built **Saral Vidhya** and **Medha** end-to-end across **TypeScript, React, Node.js, and Python**, driving features from system design and API integration through debugging and production-ready product demonstrations.
- Managed **Google Cloud Platform (GCP)** infrastructure, **Google Cloud Storage (GCS) buckets**, database hosting, and cloud service configurations to reliably store, index, and stream high-volume AI-generated media and application artifacts.
- Engineered backend-to-frontend artifact delivery pipelines and third-party API integrations (LLM, TTS/audio, storage, and database layers), maintaining consistent data flow and state synchronization across product workflows.
- Designed a multi-stage **AI video-generation pipeline** featuring act-level script segmentation, audio-transcript validation, prompt optimization, and **word-level audio/video timing alignment** for lip-sync-ready video assembly.

Founder & Lead Engineer **Aug 2025 - Present**

StartHub AI Remote

- Built and deployed a full-stack AI platform using **React, Next.js, FastAPI, and Google Cloud Firestore**, scaling to **60+ active users**.
- Architected end-to-end frontend, backend APIs, database modeling, authentication, and automated **Vercel CI/CD** production deployment workflows.

PROJECTS

ProjectBrain (projectbrain-mcp): 6D Behavioral IR, AST Context Compiler & Patch Engine | Python, Tree-sitter, MCP, SQLite, PyPI

- Published **projectbrain-mcp on PyPI**, a local-first codebase intelligence engine and **MCP server** (Claude Code, Cursor, VS Code, Antigravity) with a **SQLite WAL incremental store** indexing multi-language repos (Python, TS/JS, Go, Rust, Java, SQL) in **~90ms** via Tree-sitter.
- Designed a **Universal 6D Behavioral IR & 4D AST Coordinate Matrix (X × Y × Z × W × E × T)** with **interprocedural effect bubbling** and **cross-function state-channel graphs**, decomposing monolith files into sub-symbol cells indexed by scope, role (STATE/EFFECT/ACTION/IO/VIEW), state mutations (ΔZ), call wiring, exit contracts, and async/reactive execution modes.
- Implemented **4-Layer Multiplex Personalized PageRank (PPR)**, **Causal Symptom Chop** (tracing HTTP 4xx/5xx & SQL failures backward across frontend-to-backend hops), and a **Submodular Token Optimizer** to compile single-read **<800-line Closed-World Engineering Dossiers** with session-cache deduplication.
- Built a **Surgical Anchor Patch DSL (brain_patch)** with **<5ms Tree-sitter syntax verification** and **7-way pre-disk differential 6D contract diagnostics**, catching SQL schema/projection violations, UUID/type mismatches, transaction hazards, and cross-stack HTTP route/response shape mismatches before disk writes.
- Verified in live multi-repo A/B subagent benchmarks: reduced billed prompt tokens by **87.5% (8.02x smaller context)**, cut code-reading tokens by **81%–88%** and edit output tokens by **63%–77%**, eliminated **87%–93%** of file reads, and accelerated multi-file debugging by **2.62x** at **100% recall**.

PyPI: pypi.org/project/projectbrain-mcp **Docs:** johanabhishek.github.io/projectbrain-docs **GitHub:** github.com/Johanabhishek/ProjectBrain

MSME AI Agent: Multi-Tenant Intelligent Business Brain | Full Stack, OAuth2, LLM Orchestration

- Built a multi-tenant business intelligence platform integrating **Zoho Books via OAuth2** with an LLM reasoning layer and isolated tenant data boundaries to extract operational insights within a **36-hour GDG Agentathon Hyderabad 2025** sprint.

GitHub: github.com/Johanabhishek/MSME-Agent

MCP-Based Autonomous Commerce Agent | LangGraph, LangChain, MCP, Python

- Built a stateful autonomous commerce agent using **LangGraph, LangChain, and MCP** to orchestrate multi-step tool interactions, contextual memory across turns, and natural-language purchasing workflows.

GitHub: github.com/Johanabhishek/MCP-based-agent

Audio Deepfake Detection System | PyTorch, ResNet, GRU, Attention

- Developed a deep learning audio forensics pipeline combining **ResNet, GRU, and Attention** achieving **99.81% accuracy**; deployed inference on **Hugging Face Spaces** and authored a final-year thesis paper.

GitHub: github.com/Johanabhishek/final-voice-deepfake **Demo:** huggingface.co/spaces/pranay-ai-ml/Audio-Deepfake-Detection

EDUCATION

B.Tech in Computer Science and Engineering — JNTUA College of Engineering, Pulivendula **2022 - 2026**

Class XII (MPC): 92.5% — Krishna Murthy Junior College, Nellore 2021